EEE303 Tutorial 6b

1. A 480-V, 60 Hz, four-pole synchronous motor draws 50 A from the line at unity power factor and full load. Assuming that the motor is lossless, answer the following questions:
   (a) What is the output torque of this motor?
   (b) What must be done to change the power factor to 0.8 leading? Explain your answer, using phasor diagrams.
   (c) What will the magnitude of the line current be if the power factor is adjusted to 0.8 leading?

2. A six-pole Δ-connected synchronous motor (480-V, 60 Hz, 400-hp 0.8-PF-leading) has a synchronous reactance of 1.1 Ω. Ignore its negligible armature resistance, friction, windage, and core losses.
   (a) If this motor is initially supplying 400 hp at 0.8 PF lagging, what are the magnitudes and angles of E_A and I_A?
   (b) How much torque is this motor producing? What is the torque angle δ? How near is this value to the maximum possible induced torque of the motor for this field current setting?

   (c) What is the new I_A and power factor, if E_A is increased by 15 percent?

3. A 3-phase, unity power factor synchronous motor rated 400 hp, 2300 V, 450 r/min, 80 A, 60 Hz, drives a compressor. The stator has a synchronous reactance of 0.88 pu, and the excitation Eo is adjusted to 1.2 pu.
   a. The value of Xs and of Eo, per phase
   b. The pull-out torque
   c. The line current when the motor is about to pull out of synchronism